QP Code:25901913

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU- UGP (Honours)/MGU- BBA (Honours)/MGU- BCA (Honours)
Special Examinations, July 2026
(for 2024 Admission Regular Examinations, October 2025)
Third SEMESTER
Discipline Specific Core Course (DSC)
MG3DSCMAT202
ESSENTIALS OF APPLIED MATHEMATICS
(2024 ADMISSION ONWARDS)
Scheme For Valuation

Duration: 2 Hours Maximum Marks: 70

Part A {Short Answer Questions}
Answer any 5 questions. Each question carries 2 marks

1. Prove that $\lim_{z\to\infty}\left[\frac{2z+25}{3z-37}\right]=\frac{2}{3}$ ... [U] / [CO1]

   - $$\lim_{z\to\infty}\frac{2z+25}{3z-37}=\frac{2+0}{3-0}=\frac{2}{3}.$$

   ... 2 Marks

2. Check whether $u(x,y)=x^2+y^2$ is harmonic. ... [U] / [CO1]

   - $u_{xx}+u_{yy}=2+2=4\neq 0.$ ... 2 Marks
   - $u(x,y)=x^2+y^2$ is not harmonic ... 1 Mark

3. Find the integrating factor for the differential equation $y'+y=\frac{1}{1+e^{2x}}$. ... [U] / [CO2]

   - The integrating factor is I.F. $=e^{\int P(x)\,dx}=e^{\int 1\,dx}=e^x$. ... 2 Marks

4. Solve the differential equation $\frac{dy}{dx}=\frac{y}{x}$. ... [U] / [CO2]

   - Separate the variables: $\frac{dy}{y}=\frac{dx}{x}$. ... 2 Marks
   - Calculations to get, $y=Cx$, where C is an arbitrary constant. ... 1 Mark

5. Find any two polar coordinates of the point $(3,\frac{\pi}{4})$. ... [K] / [CO3]

   - $(r,\theta+2n\pi)$ ... 1 Mark
   - $(-r,\theta+(2n+1)\pi)$ ... 1 Mark

6. Define an ellipse. Also define foci of the ellipse. ... [K] / [CO3]

- **Ellipse:** An ellipse is the locus of a point in a plane such that the sum of its distances from two fixed points is constant. ... 1 Mark
- **Foci:** The two fixed points used in the definition of an ellipse are called its **foci**. ... 1 Mark

7. Define the unit tangent vector to a smooth curve $\vec{r}(t)$. ... **[K] / [CO4]**

   - $\vec{T}(t) = \dfrac{\vec{r}\,'(t)}{\|\vec{r}\,'(t)\|}$. ... 2 Marks

8. Find $\lim\limits_{t\to 0}\left[\cos t\,\hat{i} + \sin t\,\hat{j} - t\,\hat{k}\right]$. ... **[U] / [CO4]**

   - $\lim_{t\to 0}\left(\cos t\,\hat{i} + \sin t\,\hat{j} - t\,\hat{k}\right) = \hat{i}$ ... 2 Marks

[5 × 2 = 10]

Part B - Short Essay Questions
Answer any 5 questions. Each question carries 6 marks

9. Verify that $z = 1 - i$ satisfies the equation $z^2 - 2z + 2 = 0$. ... **[U] / [CO1]**

   - Calculations to get $z^2 - 2z + 2 = -2i - 2(1-i) + 2 = -2i - 2 + 2i + 2 = 0$. ... 5 Marks
   - Hence,$z = 1 - i$ satisfies the equation $z^2 - 2z + 2 = 0$. ... 1 Mark

10. Determine the real and imaginary parts of $f(z) = \dfrac{1}{1-z}$. ... **[U] / [CO1]**

    - Calculations to get $f(z) = \dfrac{1-x}{(1-x)^2+y^2} + i\dfrac{y}{(1-x)^2+y^2}$. ... 5 Marks
    - Real Part is $\dfrac{1-x}{(1-x)^2+y^2}$ and Imaginary part is $\dfrac{y}{(1-x)^2+y^2}$

11. Given $z_1 = 2e^{i\pi/3}$ and $z_2 = 3e^{i\pi/6}$, find the value of the product $z_1 z_2$ in the form $a + ib$. ... **[U] / [CO1]**

    - Calculations to get $z_1 z_2 = 2 \cdot 3e^{i(\pi/3+\pi/6)} = 6e^{i\pi/2}$. ... 4 Marks
    - $e^{i\pi/2} = \cos\dfrac{\pi}{2} + i\sin\dfrac{\pi}{2} = i$ and $z_1 z_2 = 6i = 0 + 6i$. ... 2 Marks

12. Find the particular solution for the differential equation $\dfrac{dy}{dx} + y\tan x = \sin x$ that satisfies the initial condition $y = 1$ when $x = \dfrac{\pi}{4}$. ... **[An] / [CO2]**

    - The integrating factor is

$$I.F. = e^{\int \tan x\,dx} = e^{-\ln(\cos x)} = \sec x.$$

... 2 Marks

    - Calculations to find $y = \cos x\left[-\ln(\cos x) + C\right]$. ... 2 Marks

- To find the required particular solution, $y = \cos x \left(\sqrt{2} - \ln\sqrt{2} - \ln(\cos x)\right)$ ... 2 Marks

13. Find the vertex, focus, directrix and equation of the axis of the parabola $x^2 = 6y$. Also, sketch the parabola. ... **[U] / [CO3]**

- Calculations to find Vertex $= (0, 0)$ ... 2 Marks
- Calculations to find Focus $= \left(0, \frac{3}{2}\right)$ ... 2 Marks
- Calculations to find Directrix $= y = -\frac{3}{2}$ ... 1 Mark
- Calculations to find $Axis = x = 0$. ... 1 Mark
- Correct Graph ... 2 Marks

Maximum Marks 6

14. Find the vertices, foci, eccentricity and directrices of the hyperbola $5x^2 - 4y^2 = 20$. ... **[U] / [CO3]**

- Calculations to find Vertices : $(\pm 2, 0)$ ... 2 Marks
- Calculations to find Foci : $(\pm 3, 0)$ ... 2 Marks
- Calculations to find Eccentricity : $\frac{3}{2}$ ... 1 Mark
- Calculations to find Directrices : $x = \pm\frac{4}{3}$. ... 1 Mark

15. Find the curvature of the curve $\vec{r}(t) = t^2\hat{i} + t^2\hat{j} + t^2\hat{k}$. ... **[U] / [CO4]**

- $$\kappa = \frac{|\vec{r}'(t) \times \vec{r}''(t)|}{|\vec{r}'(t)|^3},$$
  ... 2 Marks
- $\vec{r}'(t) \times \vec{r}''(t) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2t & 2t & 2t \\ 2 & 2 & 2 \end{vmatrix} = \vec{0}$, ... 2 Marks
- Calculations to get $\kappa = 0$ ... 2 Marks

16. The position vector of a particle at time t is given by $\vec{r}(t) = t^2\hat{i} + (2t-1)\hat{j} + t^2\hat{k}$. Find the particle's velocity, speed, direction of motion and acceleration at $t = 2$. ... **[A] / [CO4]**

- Calculations to get Velocity at $(t = 2) = 4\hat{i} + 2\hat{j} + 4\hat{k} = 6$ ... 2 Marks
- Calculations to get Speed at $(t = 2) = 6$ ... 2 Marks
- Calculations to get Direction of motion $= \frac{2}{3}\hat{i} + \frac{1}{3}\hat{j} + \frac{2}{3}\hat{k}$ ... 1 Mark
- Calculations to get Acceleration at $(t = 2) = 2\hat{i} + 2\hat{k}$. ... 1 Mark

[6 × 5 = 30]

Part C - Essay Questions
Answer any 3 questions Each question carries 10 marks

17. Consider the function $f(z) = e^{-z}$. ... [U] / [CO1]

(a) Express $f(z)$ in the form $u(x, y) + iv(x, y)$.

- Calculations to get $f(z) = e^{-x}\cos y - i\, e^{-x}\sin y$. ... 4 Marks

(b) Using Cauchy-Riemann equations, show that $f(z)$ is analytic everywhere in the complex plane.

- For proving

$$u_x = v_y, \qquad u_y = -v_x$$

are satisfied for all $x, y \in \mathbb{R}$. Since the first-order partial derivatives are continuous everywhere,

... 3 Marks

(c) Find $f'(z)$ and $f'(0)$.

- $f'(z) = -e^{-z}$ and $f'(0) = -1$. ... 3 Marks

18. Show that the differential equation

$$(1 + 4xy + 2y^2)\, dx + (1 + 4xy + 2x^2)\, dy = 0$$

is exact and solve it. ... [U] / [CO2]

- Calculations to verify

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x},$$

... 5 Marks

- For getting final solution as $(x + y)(1 + 2xy) = C$ ... 5 Marks

19. ... [U] / [CO2]

(a) Check whether the function $y = e^x$ is a solution of the differential equation

$$y'' - 5y' + 4y = 0.$$

- For the proof of $y = e^x$ is a solution of the differential equation. ... 5 Marks

(b) Verify that the function $y = C_1e^{4x} + C_2e^{-2x}$ is a solution of the differential equation $y'' - 2y' - 8y = 0$.

- For the verification of $y = C_1e^{4x} + C_2e^{-2x}$ is a solution of $y'' - 2y' - 8y = 0$.

... 5 Marks

20. Identify the following conics and find their directrix that corresponds to the focus at the origin: ... [A] / [CO3]

(a) $r = \dfrac{12}{3 + \dfrac{3}{4}\cos\theta}$

- For Calculations to get Conic is parabola , $e = 1$, Directrix: $x = 4$.
  ... 5 Marks

(b) $r = \dfrac{4}{2 - \sin\theta}$.

- For Calculations to get Conic is Ellipse ,$(e = \frac{1}{2})$, Directrix: $y = -4$. ... 5 Marks

21. (a) Find ∇f for the function $f(x, y, z) = e^x + y\cos z + (y+1)\sin x$ at the point $(0, 0, 0)$.
... **[U] / [CO4]**

- Calculations upto $\nabla f = (e^x + (y+1)\cos x)\,\hat{i} + (\cos z + \sin x)\,\hat{j} - y\sin z\,\hat{k}$.
  ... 3 Marks
- $\nabla f(0, 0, 0) = 2\hat{i} + \hat{j}$ ... 2 Marks

(b) Find the directions in which the function $f(x, y) = x^2 + xy + y^2$ increases most rapidly and decreases most rapidly at $(-1, 1)$.

- Direction of maximum increase $= \dfrac{-\hat{i} + \hat{j}}{\sqrt{2}}$, ... 3 Marks
- Direction of maximum decrease $= \dfrac{\hat{i} - \hat{j}}{\sqrt{2}}$. ... 2 Marks

22. Find the length of the indicated portions of the following curves: ... **[U] / [CO4]**

(a)

$$\vec{r}(t) = (2 + 5t)\hat{i} - (2t + 1)\hat{j} + t\hat{k}, \quad 0 \le t \le 3$$

- Length of the curve is $L = 3\sqrt{30}$ ... 5 Marks

(b)

$$\vec{r}(t) = 6\sin 2t\,\hat{i} + 6\cos 2t\,\hat{j} + 5t\,\hat{k}, \quad 0 \le t \le \pi.$$

- Length of the curve is $L = 13\pi$ ... 5 Marks

[3 × 10 = 30]